

Adaptive Traffic Light Scheduling Based On Traffic Density

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Abstract—Traffic congestion in urban areas has intensified due to the annual growth of motorized vehicles each year. In addition, traffic congestion occurs at peak hours. This situation significantly raises fuel consumption and carbon emissions to a high level, which impacts the environment, not only in direct costs but also indirect costs such as lost time, decreased productivity, and increased driver stress. To address this issue, this research proposes adaptive traffic lights using Multi-Agent Reinforcement Learning (MARL) based on the traffic density on each road section. Each intersection is modeled as an independent agent that autonomously performs decision-making processes. To prove the effectiveness, this research presents a comparison of the average waiting time, average speed of vehicles, queue length, and throughput between fixed traffic light methods and the proposed method. Through the implementation of this method, the average waiting time of vehicle was reduce 86,67 %. Traffic conditions were simulated using the Simulation of Urban Mobility (SUMO) platform, which is widely employed for microscopic traffic modeling and analysis. By improving traffic signal control through MARL, this research directly support Smart Mobility pillar of the Smart City framework, which aims to leverage intelligent systems for efficient, sustainable, and adaptive urban transportation.

Keywords—Smart Traffic Light, Reinforcement Learning, DQN, Simulation of Urban Mobility

I. INTRODUCTION

For more than half a century, traffic congestion presents significant challenges. It causes considerable time delays in transportation, which in turn increase driver fatigue, compromises safety, contributes to environmental pollution, and raises economic costs. In 2050, approximately 68% of the world's population is projected to live in urban areas, a phenomenon that has fundamentally changed the way people interact with their environment[1]. Traffic congestion has become a major issue in many cities around the world, driven by the increasing number of vehicles each year, which outpaces the development of infrastructure[2]. Traffic congestion is a daily problem that not only leads to increased fuel consumption and a higher carbon footprint, but also causing indirect losses such as wasted time, reduced productivity, and increased driver stress [3].

Current traffic light control mostly uses a fixed time the green light is allocated for a predetermined duration in every cycle without considering real traffic conditions. Consequently, traffic flow become paralyzed during peak hours [4], [5]. Traffic lights are important infrastructure in urban environments to regulate vehicles to avoid accidents and improve traffic flow. In highly congested city, prolonged vehicle stoppage leads to higher level of pollution, increased carbon dioxide emission [6], [7].

Numerous studies have been conducted on traffic light control optimization and environment impact using various approaches [8] including dynamic control system based on queue theory [9], centralized traffic light control [1] and

estimate queue length and traffic flow using V2I data [10], using fuzzy inference methods [2], [11], Software Defined Vehicular Network [12], Reinforcement Learning method with various modifications [13], [14], [15]. The use of deep reinforcement learning to Traffic Signal Control System (TSCS) has been the subject of extensive research in recent years. For example, a TSCS employing a deep Q-network (DQN) has been employed to lessen traffic congestion in an environment with a single intersection. However, TSCS are supposed to communicate with one another in real-world settings with multiple intersections. In recent developments, the growth of Artificial Intelligent (AI) has significantly facilitated for modernization and intelligent manufacturing. Among various AI techniques, RL has emerged as key approach for solving complex decision-making task. It has applicable across multiple domains, including autonomous driving, robotic, game development, and resource management[16].

While previous research has primarily addressed traffic control at individual intersections without accounting for emergency vehicle prioritization, this study expands the scope by focusing on multi-intersection scenarios and explicitly integrating emergency vehicles as a critical factor in signal optimization. This research proposed, Multi-Agent Reinforcement Learning to dynamically adjust traffic signal control based on current traffic conditions. The approach aims to optimize traffic light duration in multiple intersections, which each intersection contributes to managing of vehicle speed based on traffic density, active phase, average waiting time, and the presence of emergency vehicles such as ambulances and fire trucks. Using the MARL method is expected to reduce the average travel time, vehicle queue length, and improve traffic flow. The proposed method evaluated using Simulation of Urban Mobility (SUMO), a traffic microsimulation tool equipped with an API appropriate for researchers. SUMO allows model to be tested under conditions that closely resemble real-world urban traffic environments.

II. LITERATURE REVIEW

The original traffic signal system was based on priority order, predetermined cycle length, and phase division. The method was adopted by current traffic systems in the world, and works well under fixed traffic based on prior knowledge from experts. However, the system cannot handle dynamic traffic, thus requiring human assistance to fine-tune the system to cope with traffic changes [17]. There are several studies related to adaptive traffic light signal control:

Hou [9], developed an ideal hybrid model for discrete traffic signals and continuous traffic flow coordination along urban artillery roads, the study assumed that traffic as a platoon and focused on the coordination of different

intersection, however the study did not consider emergency vehicles to be given priority at intersection and dynamic traffic conditions.

Koch [8], proposed an adaptive traffic light control system using Proximal Policy Optimization (PPO), a model-free DRL algorithm. Unlike traditional methods, their approach integrates microscopic traffic simulation (SUMO) with physics-based powertrain models for both Internal Combustion Engine Vehicles (ICEVs) and Electric Vehicles (EVs). Their findings reveal that the DRL-based adaptive control significantly improves average vehicle speed (by 41%) while maintaining or slightly reducing total CO₂ emissions.

Yanan [14], offer an innovative framework for spatiotemporal multi-agent reinforcement learning (STMARL) that effectively coordinates traffic light control and captures the spatiotemporal relationships of several intersections, however this research was not include emergency vehicle as priority in traffic light. In other research of Reinforcement Learning research [18], [19] implemented in multi intersection but not prioritize the emergency vehicle, Liu [20], only use single intersection environment and did not prioritize emergency vehicle.

Sachan [21], proposed traffic signal control using IoT and edge computer infrastructure. The proposed framework features a heuristic driven, actuated traffic signal control built on S-Edge, which integrated with Fuzzy Inference System (FIS) to determine phase-cycle length at intersection. The FIS operates on a fog computing node, while a lightweight algorithm is used to assess lane pressure. Despite its strengths in real-time responsiveness, the framework primarily designed for single-intersection.

Recent advancements in traffic light control have demonstrated notable progress, with decision-making systems adaptively and learn based on real-time traffic conditions. however, prior research has predominantly concentrated on single or multi-intersection scenarios without incorporating mechanisms to prioritize emergency vehicles at traffic signals. To address this limitation, we propose a novel approach utilizing Multi-Agent Reinforcement Learning (MARL) across multiple intersections, explicitly integrating emergency vehicle prioritization into the adaptive control framework.

III. PROPOSED METHOD

Researchers applied the Multi Agent Reinforcement Learning (MARL) technique to four traffic intersections in order to create the Smart Traffic Light. Each traffic light functions as an agent, providing data such as phase active, traffic density, average waiting time, and emergency vehicle.

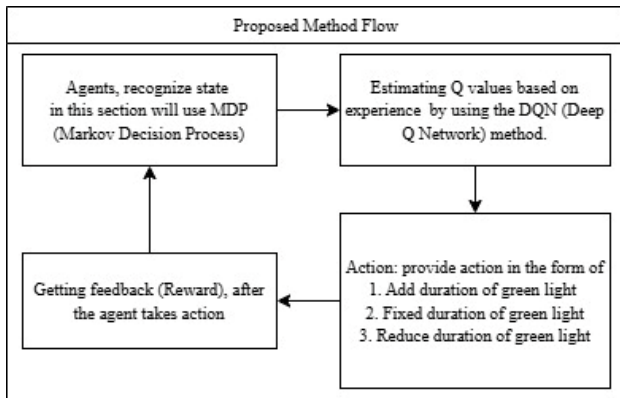


Fig 1. Proposed method flow

A. Reinforcement Learning

Reinforcement Learning (RL), which is part of machine learning, is a machine learning approach where an agent interacts with the environment [4] and performs trial and error actions through learning experiences in the local environment with the aim of increasing the maximum reward[13], [22], [23]. Figure 1 shows algorithm of RL, first step, RL can be formulated under the Markov Decision Process (MDP) framework which can be used to learn Adaptive Traffic Signal Control (ATSC) based on real traffic conditions [24]. Mathematically, MDP as a control process provides a framework, presenting complex decision making in a non-deterministic environment. The interaction between the agent and the environment is described as state, action, reward [25]. The agent and environment interact in discrete time step, and send reward to the agent[13]. The ultimate goal of the agent is to take action according to the given scenario[4]. The following is an explanation of the three components of RL:

- State is utilized to give the RL agent changing environmental input at each step, it has a significant impact on the RL agent's actions and performance to continuously supply the RL agent with shifting environmental inputs. In this study, define the number of states, including the active phase, average waiting time, traffic density, and emergency vehicles.
- Action, the agent will choose the action to be performed. At this stage, the agent will do some action
 - Add duration of green light
 - Fixed duration, its mean no action to do
 - Reduce duration of green light.
- Reward, shows how well or poorly the agent accomplished the action. The reward shows how well or poorly the agent accomplished the action. As a directional signal, reward will push the Q value down for actions that make worse situation and up for actions that create best situation (such as smooth traffic and emergency services).

The state of an environment at time step t is denoted s_t . Thus the agent examines s_t and performs the corresponding action a_t . Then the environment changes the state s_t to s_{t+1} and provides a feedback reward r_{t+1} to the agent. In this process, the agent's decision is formalized by defining a policy. The mapping function of each state s to be taken from that state is called policy π . Policy is deterministic if choosing action a from s : $p(a|s) = 1$, otherwise, it is stochastic if state s : $p(a|s) < 1$. In that case policy π is defined as a probability distribution of candidate actions formulated as follows

$$\pi = \Psi(s) = \{p(a_i|s) \mid \forall a_i \in \Delta_\pi \wedge \sum_i p(a_i|s) = 1\} \quad (1)$$

Where Δ_π represents all candidate actions of the policy [13]. It is assumed that the action space is discrete so that it can be summarized with integral notation [26].

B. Multi Agents Reinforcement Learning (MARL)

In ATSC, multiple signalized intersection agents have a stake in optimizing global network traffic. The traffic conditions at each intersection determine the actions to be taken by each agent. Each intersection act as an independent agent that learns and makes decisions based on traffic condition in its local area, collectively contribute to all traffic network efficiency. MARL, mostly formulated with Q-Learning, distributes the global action to each local agent by assuming a global decomposable function Q.

Each traffic intersection is modeled as autonomous agent within a Multi Agent Reinforcement Learning. The learning process of agent is using Q-learning algorithm, as represent by following equation:

$$Q_i(s_t^i, a_t^i) \leftarrow Q_i(s_t^i, a_t^i) + \alpha [r_t^i + \gamma \cdot \max_{a'} Q_i(s_{t+1}^i, a') - Q_i(s_t^i, a_t^i)] \quad (2)$$

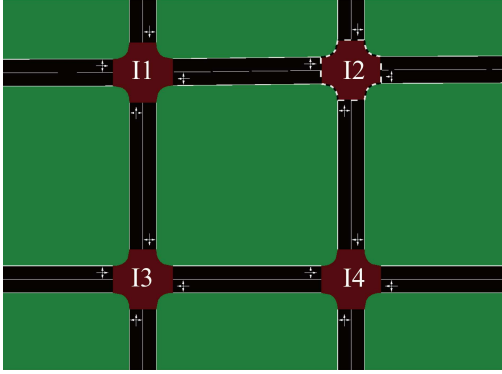


Fig 3. Road structure with 4 intersections

$$Q_i(s_t^i, a_t^i) \leftarrow Q_i(s_t^i, a_t^i) + \alpha [r_t^i + \gamma \cdot \max_{a'} Q_i(s_{t+1}^i, a') - Q_i(s_t^i, a_t^i)] \quad (2)$$

The equation describes $Q_i(s_t^i, a_t^i)$ as Q-value function for agent i at time t , which represent the expected cumulative reward when taking action a_t^i in state s_t^i . The update is influenced by learning rate α , which control how quickly the agent adapt, and discount factor γ , determines the importance of rewards. r_t^i represents the reward received by agent i after executing the action, the value based on local traffic performance metrics, such as average waiting time, queue length, and average speed of all vehicles. $\max_{a'} Q_i(s_{t+1}^i, a')$ identifies the best expected return from the next state s_{t+1}^i . Through this process, each agent learn to select traffic signal actions, such as increasing, decreasing or maintaining green light duration.

C. Deep Q-Network (DQN)

Deep RL is a term that denotes the combination of deep learning and reinforcement learning to handle high-dimensional environments[15]. DQN consists of two important components, the target network and the experience repetition implementation[18]. By employing DQN algorithm, complex problems can be solved and this is proven in classic Atari games, AlphaGo, MuJoCo physical problems [19]. This algorithm use a neural network that approximates the Q function (action value) by accepting input in the form of environmental conditions (state) and producing output in the form of Q values for every potential action.

When given a certain state, it produces a vector denoted as $Q(s, a; \theta)$, where θ denotes the network parameters. The target reflects the evaluation network, which is characterized by the parameter θ^- . However, the matching parameters of the evaluation network are copied to update the parameters for each step τ . Update parameter is formulated below:

$$L_k(\theta_k) = E_{(s,a,r,s')} \left[\left(r + \gamma \max_{a'} Q(s', a'; \theta_k^-) - Q(s, a; \theta_k) \right)^2 \right] \quad (3)$$

Figure 2 illustrates the layered architecture of a deep Q-network (DQN). The states used as inputs for the DQN process include phase active, average speed, traffic density,

and the presence of emergency vehicles. Each hidden layer contains 128 neurons and employs a leaky Rectified Linear Unit (Leaky ReLU) as the activation function. The fully connected layers produce Q-values corresponding to the possible actions an agent can take. The parameters of the deep

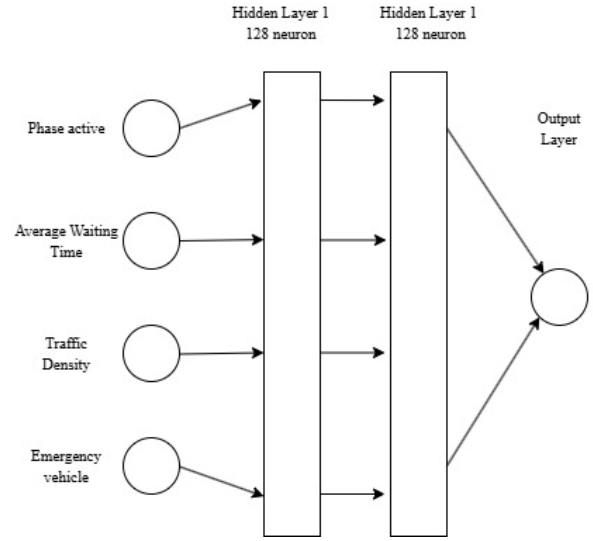


Fig 2. Neural Network of DQN

neural network are initialized with random weights. At the start of each episode, the agent evaluates the simulated traffic environment. Vectors representing phase active, average waiting time, traffic density, and emergency vehicles are extracted to define the state space for a specific intersection. Using these inputs, the agent selects the action associated with the highest Q-value. Following the action, the agent receives a reward based on the effectiveness of its performance.

D. Simulation Setup

This study utilizes synthetic traffic data that generated using the Simulation of Urban Mobility (SUMO) environment to model and evaluate traffic conditons at a four intersection network. The dataset includes 11 route variations for private vehicle and 3 predefined emergency vehicles. Private vehicle flows were defined through probabilistic flows (5 - 8 % probability per route).

Intersection name of traffic light is shown in figure 3. Each intersection consist of four directions and each direction consist of two lanes. The simulation follows Indonesia traffic rules, which vehicles drive on the left side of the road. Each intersection is controlled by an eight-phase traffic signal system shown in figure 4 with a three-second yellow phase during transition from green to red light. Two types of vehicles are used in this research, the first type is general

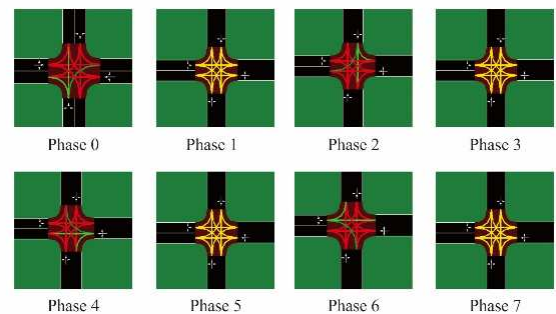


Fig 4. Phase of traffic light

vehicles with probabilities ranging from 0.05 to 0.08 at each edge and second is two emergency vehicles that will appear at 100 and 300 step simulations. The traffic flow entering each intersection possesses varying probability distributions, so resulting more diverse traffic conditions, where simulation conducted for 1800 simulation steps that each step represent one seconds of real world time.

E. Metrics Evaluation

To evaluate the proposed method, researchers used the metrics of average vehicle waiting time, average vehicle speed, vehicle queue length, and vehicle throughput. Then it will be compared with the fixed traffic light method which is generally used in various intersection.

IV. RESULT AND DISCUSSION

The experimental results of the proposed Multi-Agent Reinforcement Learning (MARL) method compared with the Fixed Time Traffic Light method are analyzed in this section. To validate the proposed MARL approach, the researchers conducted extensive traffic simulations using Simulation of Urban Mobility (SUMO).

As previously mentioned, for optimal traffic light regulation, RL shows "superior performance over conventional methods". Figure 5 show that top graph is average waiting time per intersection, middle graph is average speed of vehicle, and bottom graph is queue length of proposed each intersection. Each intersection is represented by a specific color for clarity.

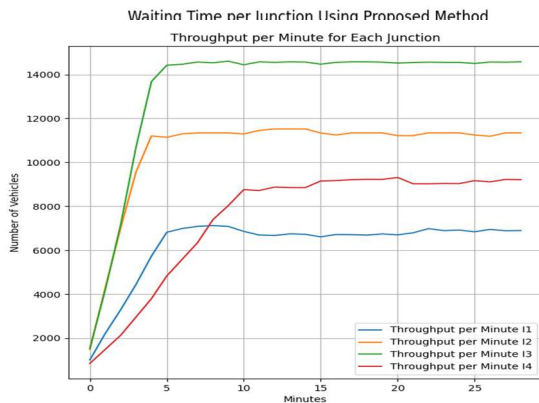


Fig 7. Throughput per minute for each intersection under Proposed MARL Method

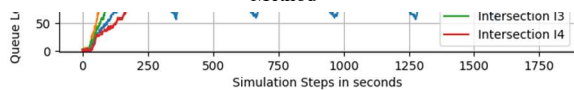


Fig 5. Cumulative average waiting time, average speed, queue length with proposed method traffic light

From average waiting time graph can be observed that, the maximum number of cumulative average waiting time reached approximately 4,000 seconds at intersection I3 (green line) and the minimum one was observed at intersection I1 (blue line), approximately 2,000 seconds after 1,800 step. Regarding vehicle speed, the data shows a range between 0 m/s to 15 m/s (0 Km/h), the highest average speed was achieved at intersection I4. The last, the bottom graph show that average queue length reached 200 meters at intersection I3 and the shortest queue was recorded at intersection I1.

For comparison, figure 6 presents the results of the fixed time traffic light method. The cumulative average waiting time surpassed 30,000 seconds at Intersection I4, indicating that vehicles had to wait significantly longer compared to the

MARL method. The queue length for the fixed method exceed 300 meters, particularly at intersections I3 and I4.

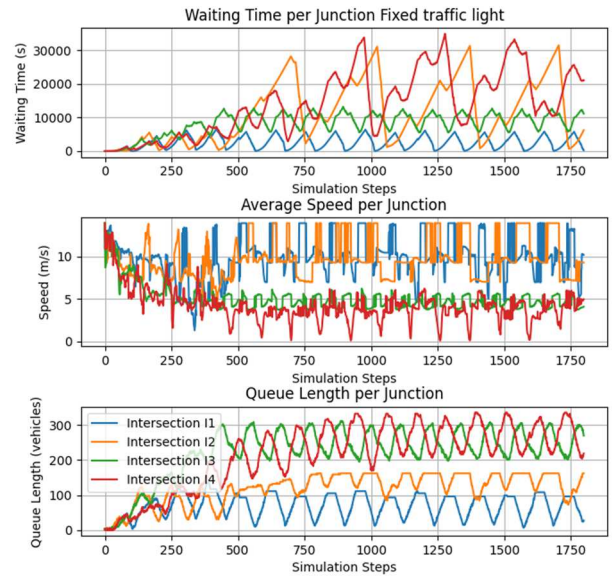


Fig 6. Cumulative average waiting time, average speed, queue length fixed time method

Although the average vehicle speed using MARL is slightly lower than fixed time method, this slight reduction results from more dynamic and frequent traffic signal changes, which prevent prolonged stopping but required moderate deceleration, thereby maintaining smoother traffic flow.

Furthermore, the throughput results were also analyzed to evaluate the intersection efficiency. Figure 7 shows the throughput results for the MARL method. The system achieved a stable and high throughput across all intersections after the initial learning phase. Intersection I3 achieved the highest throughput, reaching approximately 14,500 vehicles per minute, followed by Intersection I2 with about 11,000 vehicles per minute. Throughput values remained stable, reflecting the MARL method's ability to adapt to changing traffic conditions. In comparison, Figure 8 presents the throughput results for the Fixed Time Traffic Light method. The throughput exhibited significant fluctuations at all intersections, with no clear stabilization throughout the simulation. Intersections I3 and I4 showed particularly erratic throughput patterns, confirming the inefficiency of fixed timing under dynamic traffic conditions.

V. CONCLUSION

This research proposed an adaptive traffic signal control system utilizing Multi-Agent Reinforcement Learning (MARL) to address the inefficiencies inherent in conventional Fixed Time Traffic Light systems. By modeling each intersection as an independent intelligent agent and incorporating real-time variables such as traffic density, average waiting time, and the presence of emergency vehicles, the system dynamically adjusted signal timings to optimize traffic flow.

From results of two method, MARL method significantly improve traffic efficiency. Average waiting time of MARL consistently shorter than fixed time, almost 86,67% reducing that occurs in fixed time and reducing the queue length of vehicle until 33,33 %. Although the average speed of vehicles in the RL method is slightly lower than the fixed time, this condition can be attributed to the dynamic adjustment of the signals, which requires vehicles to slow down frequently, but

does not make vehicle stop for long time. In conclusion, the MARL-based adaptive traffic light system successfully enhances urban traffic efficiency, reduces congestion, and adapts dynamically to varying traffic conditions. By quantitatively demonstrating improvements across all key performance indicators, this research substantiates the value of reinforcement learning for smart urban mobility initiatives. Future extensions of this work could involve scaling the model to larger, more complex traffic networks and integrating additional real-world variables to further refine system robustness and applicability.

ACKNOWLEDGMENT

The author would like to thank the Ministry of Communication and Digital of the Republic of Indonesia for providing support through scholarships and facilitating activities during the study period.

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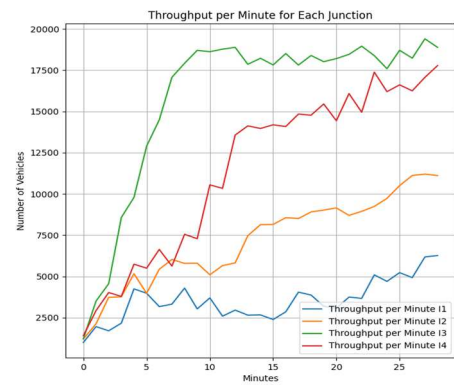


Fig. 8 Throughput per minute for each intersection under Fixed Time Method.

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